ORDINAL DEEP LEARNING FOR AUTOMATED FALL ARMYWORM DAMAGE SEVERITY SCORING IN FAWCHECK: A CNN-BASED APPROACH TO THE PRASANNA SCALE

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ABSTRACT

GIOVANN JOSH N. APOLINAR, University of the Philippines Los Baños, NOVEMBER 2026. **ORDINAL DEEP LEARNING FOR AUTOMATED FALL ARMYWORM DAMAGE SEVERITY SCORING IN FAWCHECK: A CNN-BASED APPROACH TO THE PRASANNA SCALE**

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(abstract text here)

INTRODUCTION

Background of the Study

The fall armyworm (FAW), *Spodoptera frugiperda* (J. E. Smith), is among the most invasive pests that bring serious economic loss to Philippine maize production. The Department of Agriculture recorded infestations affecting more than 5,000 farmers across approximately 3,200 hectares by mid-2024, as reported in the FAWCheck v1 study (Casasola, Anacleto, & Nocum, 2026). By mid-August 2024, the pest had destroyed an estimated 734 metric tons of corn worth about PHP 57.03 million. These losses contributed to a national deficit, with second-quarter output falling 20.35%. Farm grain prices, meanwhile, remained nearly flat while retail corn prices rose 5.75%, leaving growers selling less grain of lower quality with no price gain to offset the loss. Because of the FAW's ability to spread quickly, the time between detection and intervention can often mean the difference between a manageable infestation and a lost crop. Effective integrated pest management necessitates the ability to determine not only whether a plant is infested, but also the extent of damage the plant has sustained, as varying degrees are important in evaluating germplasm resistance.

Typically, the screening process uses a visual rating scale as the basis for evaluation. The scale used in this study is the one to nine foliar damage scale of Prasanna, Huesing, Eddy, and Peschke (2018), which is modified from the whorl-stage screening scales of Davis, Ng, and Williams (1992). A field technician assesses the plant, matches what is visible against a set of ordered descriptors, and assigns an integer score.

Thus, the main motivation of this study lies in two characteristics of this rating scale. First, the scale is ordinal and not nominal. A score of 5 denotes a significantly greater degree of damage than a score of 4. A model that treats the nine categorical scores as unordered labels discards this ranking structure. With the use of multi-class cross-entropy,

predicting 3 when the true target is 6 incurs the same loss as predicting 5 when the true target is 6. Essentially, such a loss function is blind to order, which is fundamental in this problem.

Second, the distance between scores carries meaning in the screening process. Prasanna et al. (2018) present the scale for assessing foliar damage in maize germplasm entries, and group its scores into resistance-response bands: score 1 highly resistant, scores 2 and 3 resistant, scores 4 and 5 partially resistant, scores 6 and 7 susceptible, and scores 8 and 9 highly susceptible (see Table 1). In assessing the resistance of maize, a confusion between 6 and 7 leaves an entry in the susceptible category, while a confusion between 5 and 7 moves the resistance label to a different categorical response, from partially resistant to susceptible. Therefore, errors on this scale are not interchangeable. A model that is blind to these differences is likely to produce inefficient results regardless of how accurate its metrics appear.

Moreover, FAWCheck v1 was built as a field monitoring tool for agricultural technicians and researchers (Casasola et al., 2026). In this use, the score describes how badly a crop is damaged. The most expensive error is scoring a susceptible or highly susceptible plant (scores 6 through 9) below 6, because it understates damage where damage is greatest. This study therefore treats scores 6 through 9 as *severe damage*, starting where the susceptible category of Prasanna et al. (2018) begins. Over-scoring also has a cost, because it overstates damage. The study measures both whether models keep plants in the correct response category, in either direction, and whether they detect severe damage (see *Reported Metrics*).

Statement of the Problem

FAWCheck v1 is an existing application that digitizes this rating workflow. It is a progressive web application which uses a convolutional neural network (CNN) to score an image of maize, along with a rule-based scoring system that feeds on a symptom

questionnaire. The outputs of the two components are fused into a final severity estimate. It is a working system with measured performance, and the present study builds on this foundation rather than displacing it. The CNN was reported at 50.00% exact accuracy, 0.4105 macro F1, 76.47% within-one-level accuracy, and 0.8088 mean absolute error (MAE). Moreover, the fused system reached 83.13% exact agreement with expert scores, 0.27 MAE, and 0.7881 weighted Cohen's kappa (Casasola et al., 2026).

However, three problems limit what can be built on those results. First, the v1 CNN treats severity scoring as multi-class classification with sparse categorical cross-entropy (Casasola et al., 2026). It cannot represent the inherent order of the severity scale, and its loss penalizes the categories with similar magnitude without accounting for the real distance among target scores. Nonetheless, the errors were already largely adjacent to the true target, as 76.47% of its predictions fell within one level of the reference (Casasola et al., 2026). This implies that the errors did not arise from random noise but are structured around the true label. Thus, the question is not whether ordinal structure matters, but whether modeling it reduces the more distant errors that remain.

Second, the v1 results cannot serve as a baseline. The CNN metrics were computed on a small evaluation set of 68 images. Applying the Wilson score method to the published counts gives a 95% confidence interval (CI) of [0.3844, 0.6156] for the CNN accuracy, an interval width of 0.2312. These findings do not point to a defect in the model, but to a limit of what a sample of 68 images can establish. At that sample size, the 95% CI spans roughly 38% to 62%, wide enough to contain both a poor classifier and a good one. Consequently, any later model that scores within this range cannot be said to have improved on the 50% baseline, since the difference would be indistinguishable from sampling noise at this evaluation size.

Third, the v1 study correctly attributes the weak performance of the CNN to class imbalance, limited samples at some severity levels, variation in field image quality, and visual overlap between neighboring categories (Casasola et al., 2026). Indeed, the dataset

is severely skewed toward the lower-end scoring categories, but this is not the only factor that contributes to the weak performance of the model. A review of the file-level structure of the train/validation/test partition, conducted for this study, shows that the partition is non-representative (see *Why the Inherited Partition Is Discarded*). Because validation results also guided model selection in v1 (Casasola et al., 2026), that decision process inherited the same bias from the non-representative partition. This is a separate defect from class imbalance, and any before-and-after comparison run on this partition measures artifacts of the partition rather than genuine differences in model performance.

Objectives

To determine whether ordinal regression improves automated FAW foliar damage severity scoring on the Prasanna et al. one to nine scale relative to standard multi-class classification, under an evaluation protocol capable of supporting that comparison.

Specifically, the researcher would like to do the following:

1. To compare the rank-consistent CORAL (Cao, Mirjalili, & Raschka, 2020) and CORN (X. Shi, Cao, & Raschka, 2023) and the regression-based THOR (Fuchs & Keshet, 2022) against a softmax multi-class baseline on identical cross-validation folds, with identical imbalance handling (Kang et al., 2020) applied to all four models, and determine whether ordinal regression reduces severity-scoring error.

2. To determine whether the three ordinal methods differ from one another on this task, and whether the published expectations—CORN advantaged on mean absolute error and CORAL on outlier-sensitive error (X. Shi et al., 2023), measured here as macro-averaged mean squared error (see *Evaluation Methods*), and THOR advantaged over CORAL on mean absolute error (Fuchs & Keshet, 2022)—hold under severe class imbalance at nine ordinal levels.

3. To measure the agreement between the author's and an expert's independent scores

on every newly labeled image, to locate the model's agreement with the expert relative to it, and to check the consistency of the inherited labels across near-duplicate images.

4. To rebuild the FAWCheck evaluation protocol on a stratified, leakage-controlled foundation, and determine how much of the change in measured performance is attributable to the protocol rather than to the model, by training the same v1 recipe on the same images under the inherited partition and under the rebuilt protocol.

The third objective sets the models beside a human rater: each model's agreement with the expert is reported next to the author's, on the same images (see *Human Reference*). This is a reference point, not a ceiling (see *Scope and Limitations of the Study*).

Significance of the Study

For the Institute of Plant Breeding, the Entomology Laboratory Division, and the users of FAWCheck, this study produces a foliar damage severity model whose performance is reported with quantified uncertainty: confidence intervals on accuracy, macro-MAE, and weighted kappa, taken from repeated stratified cross-validation with corrected variance and from bootstrap resampling by near-duplicate cluster. The v1 study reported its performance as a point estimate, an accuracy of 50.00% on 68 evaluation images, with no interval. This quantified uncertainty is the precondition for two decisions. First, it indicates whether the image-based score can be trusted in the field. Second, it indicates whether future dataset growth actually improves the model, which can be established if a future result clears the interval of the earlier one.

For agricultural computer vision, the study applies rank-consistent ordinal regression to an expert-rated agronomic damage scale. As discussed in *CNN Approaches to Plant Disease and Pest Severity*, existing crop severity studies mostly use a small number of coarse bins evaluated with plain accuracy. Meanwhile, ordinal methods and

agreement-based metrics are established practice in medical image grading.

Methodologically, the study contributes a reusable evaluation protocol, described in *Evaluation Protocol*, that can serve subsequent FAWCheck model iterations and other rating-scale tasks in the Laboratory's work. The protocol extends the v1 study, whose own recommendations called for expanded, better-balanced data (Casasola et al., 2026).

Scope and Limitations of the Study

The study only covers the image-based severity model of FAWCheck. It does not address cob or ear damage, yield loss, the rule-based heuristic scoring, the fusion of image-based and rule-based scores, or the integration of the resulting model into FAWCheck's deployed inference service, including its export to the ONNX format. Whether fusion improves on the image model alone is recorded as an open question but is not tested. The v1 comparison of its CNN-only and fused-system figures (Casasola et al., 2026) cannot settle it: the two figures come from different datasets, class mixes, and reference labels, and the rule engine's symptom answers mirror the descriptors behind the reference score. Settling it requires image-only, rule-only, and fused arms on the same held-out cases. The scope only covers foliar damage on the Prasanna et al. one to nine severity scale (Prasanna et al., 2018).

Moreover, the model comparison is limited to a softmax baseline and three ordinal methods: CORAL and CORN from the rank-consistent binary-decomposition family, and THOR from the regression-based family. It does not cover the whole ordinal regression family. Soft ordinal labels are excluded, and the regression-based family is represented by THOR alone, not also by plain regression with tuned cut-points (see *Scope of the Comparison, and Why THOR*). Replacing MobileNetV3Small with a stronger self-supervised backbone, such as the distilled DINOv3 ViT-S/16 (Siméoni et al., 2025), is left to future work.

Furthermore, the prediction target is an image-based Prasanna rating: a score

assigned by a rater looking at a photograph, not at the plant. The descriptors count affected leaves and refer to the whorl and furl, which a single photograph may show only in part, so both the labels and the agreement measured in this study describe rating photographs, not scoring plants in the field. The scale was also designed for maize germplasm entries, typically assessed under controlled infestation (Davis et al., 1992; Prasanna et al., 2018), whereas much of the imagery here comes from farmer fields and public datasets. FAWCheck applies the scale to field monitoring, an extension of its stated purpose, so the response categories are read as levels of damage, not as judgments of a plant's resistance. Because the scale is the derived form identified in the review of rating scales (see *Fall Armyworm Damage Rating Scales*), results reported here are not strictly comparable to published screening results reported against the original Davis or Williams scales.

Whether the ordinal comparison is tested as a confirmatory hypothesis depends on a sensitivity check made before any ordinal model is trained (see *Sensitivity Check Before the Model Comparison*). If the comparison cannot detect the smallest effect of interest even after the pre-registered responses, it is reported as exploratory.

Score 9 is represented by four images in the inherited dataset. The expansion target requires it to reach at least 15; if it does not, it is merged with score 8 for reporting under a pre-registered rule and is out of scope for quantitative per-level claims (see *Pre-Registered Merge Rule*). Even at 15 images, per-level claims for score 9 carry wide intervals.

In addition, model performance depends on the representativeness of the training data. The inherited dataset mixes field photographs with publicly sourced images, and the candidate expansion pool is drawn from a single region and season in Tanzania, captured largely on non-phone cameras. Generalization to Philippine field conditions is therefore measured, not assumed, on a Philippine external test set of locally collected phone photographs (see *Philippine External Test Set*). At a minimum of 150 images, a proportion near 50% on that test set carries a 95% confidence interval about 16 percentage points wide, against 23 for v1's 68-image evaluation.

Finally, no human ceiling is measured. Agreement is measured between the author and one expert, which partly reflects how well the author has learned the rubric and is therefore a reference point rather than a bound on achievable performance (see *Expert Annotation, Agreement, and Label Consistency*). A ceiling would need two experts scoring the same images independently, which this study does not do and the literature does not report for this scale (see *Observer Variability on the Rating Scale*). The inherited 2,084 labels, whose provenance is undocumented (Casasola et al., 2026), are not re-scored. Their quality is assessed only through their consistency across near-duplicate images, which can reveal inconsistency but cannot establish accuracy.

REVIEW OF RELATED LITERATURE

This chapter reviews the rating scale and its reliability, prior image-based approaches to plant damage and disease severity, ordinal regression methods for deep networks and their use in medical grading, and the evaluation methodology under which model comparisons on imbalanced ordinal data can be defended. It closes by stating the gap the present study addresses.

Fall Armyworm Damage Rating Scales

The 1–9 foliar damage scale used by FAWCheck derives from the visual rating scales developed by Davis et al. (1992) for screening whorl-stage maize for FAW resistance, and reaches this study through Prasanna et al. (2018). The descriptors are cumulative in character: lower scores describe shot-hole feeding on a few leaves, middle scores introduce elongated lesions of increasing extent, and upper scores describe lesions across most leaves accompanied by large irregular holes, terminating at near-total destruction of the whorl and furl.

Toepfer et al. (2021) review the family of FAW leaf damage rating scales and draw a distinction that matters for this study: Davis' 0–9 scale assesses the whorl and furl from a top view, whereas Williams' 0–9 scale assesses the whole plant. The two are frequently integrated in applied work, and Toepfer et al. observe that the scales have been repeatedly modified, and at times misapplied or miscited (Toepfer et al., 2021). They propose simplified, instar-independent variants on the grounds that the descriptive distinctions in the original scales are difficult to apply consistently in the field.

Scale identification. The operative scale for this study is the one published as Table 3 of Prasanna et al. (2018) and condensed in Table 1. Its source line reads "modified from Davis and Williams (1992)" (Davis et al., 1992). It is therefore neither Davis'

whorl-and-furl top-view scale nor Williams' whole-plant scale in its original form, but a derived scale drawing on both, and it is named as such here rather than claimed to correspond strictly to either predecessor.

Table 1. Prasanna scale for foliar damage by Fall Armyworm, with descriptors

Score	Damage symptoms	Response⋆
1	No visible leaf-feeding damage	Highly resistant
2	Few pinholes on one or two older leaves	Resistant
3	Several shot-holes on a few leaves (fewer than 5) and small circular holes	Resistant
4	Shot-holes on several leaves (6–8), or small lesions and pinholes, plus a few small elongated lesions (up to 1.3 cm) on whorl and furl leaves	Partially resistant
5	Elongated lesions (over 2.5 cm) on 8–10 leaves, plus a few small to mid-sized irregular holes eaten from the whorl or furl	Partially resistant
6	Several large elongated lesions and/or large irregular holes on several whorl and furl leaves	Susceptible
7	Many elongated lesions of all sizes on several whorl and furl leaves, plus several large irregular holes	Susceptible
8	Many elongated lesions of all sizes on most whorl and furl leaves, plus many mid- to large-sized irregular holes	Highly susceptible
9	Whorl and furl leaves almost totally destroyed; plant dying from extensive leaf damage	Highly susceptible

Adapted from Table 3 of Prasanna et al. (2018), whose source line reads "modified from Davis and Williams (1992)." ⋆Response labels are read in order from the printed column, whose entries are offset from their rows. This reading assigns score 1 to highly resistant, consistent with other descriptions of the scale; it is to be confirmed against a clean copy of the guide. Figure A.1 (Appendix A) shows an example image for each score.

The published table supports reading it as admitting both assessment framings. First, its descriptors mix whole-plant leaf counting with whorl-specific detail: scores 2 through 5 are defined by how many leaves carry injury ("1–2 older leaves," "a few leaves (<5 leaves)," "several leaves (6–8 leaves)," "8–10 leaves"), while scores 4 through 9 add the condition of the whorl and furl leaves. Second, the nine reference photographs accompanying it (Figure 5 of Prasanna et al., 2018) show both framings, some filling the frame with one plant's upper leaves and others showing whole plants in a stand. An image is accordingly taken

to satisfy this scale's assessment conditions if it shows the extent of leaf injury across the plant, the condition of the whorl and furl, or both, in either framing. This is the sense in which the capture protocol's requirement of one primary maize plant per photograph, with the whorl and upper leaves in frame, is consistent with the scale.

Observer Variability on the Rating Scale

Toepfer et al. (2021) document observer bias on these scales qualitatively, including difficulty in the mid-to-upper range, but report no kappa, intraclass correlation, or inter-rater correlation coefficient. A partial quantitative estimate does exist. In proposing a simplified TNAU 1–5 scale, Srinivasan et al. (2022) report that when the same subjects scored the same plots, the mean standard deviation across raters ranged from 0.48 to 0.80 under the Davis 1–9 scale, against 0.19 to 0.40 under the 1–5 alternative. The authors attribute the difference to overlapping descriptions between adjacent Davis scores—noting in particular that damage on one-half of leaves (score 7) and on two-thirds of leaves (score 8) is difficult to distinguish reliably—and observe that reproducibility declines as the number of scale points increases (Srinivasan et al., 2022).

Consequently, the claim that the literature supplies no quantitative estimate of rater variability on this scale is too strong; a dispersion figure exists, even if an agreement coefficient does not. However, the figure that exists is not the figure this study needs. A standard deviation across raters scoring plots is not a quadratic weighted kappa between raters scoring individual images, and the imagery, raters, and conditions differ. The TNAU result is therefore best used as a prior expectation against which this study's own measurement is positioned, not as a substitute for making it. Its finding that mid-to-upper-range distinctions are least reliable also predicts where author–expert disagreement in this study should concentrate, which is a testable expectation rather than an assumption.

A terminological caution follows from the same source: it distinguishes the original

Davis 1–9 scale from a modified version it attributes to Ni et al. (as cited in Srinivasan et al., 2022). Together with the Davis/Williams distinction above, the phrase "the 1–9 scale" is ambiguous across at least three referents, which is why the review of rating scales above names the particular derived scale this study uses. It also means that published results reported against "the Davis scale" or "the Williams scale" are not strictly comparable to results reported here, a limitation recorded in the *Scope and Limitations of the Study*.

CNN Approaches to Plant Disease and Pest Severity

Deep learning for plant disease has been dominated by identification rather than grading. Mohanty, Hughes, and Salathé (2016) and Ferentinos (2018) established that CNNs classify disease *type* from leaf imagery with high accuracy, and subsequent reviews document the breadth of that work (Boulent, Foucher, Théau, & St-Charles, 2019; Lu, Tan, & Jiang, 2021). Severity assessment is a much smaller literature: a survey of CNN-based plant disease severity assessment identifies only sixteen studies and notes that severity work remains sparse relative to identification (T. Shi et al., 2023).

The severity studies that exist share a structural pattern relevant here. They discretize severity into a small number of coarse bins—four stages for maydis leaf blight of maize (Haque et al., 2022), four to six stages for tomato bacterial spot and mosaic virus (Astani & Hasheminejad, 2026), and five bands defined by the proportion of diseased leaves in work covered by the survey (T. Shi et al., 2023)—and they evaluate with plain accuracy and F1. The bins are typically derived from measured lesion or leaf area rather than assigned by an expert against a descriptor rubric. The survey reports that the studies it reviews are evaluated with accuracy, precision, recall, mean average precision, and F1, and it describes none that uses an ordinal-aware loss or an ordinal error measure such as MAE or quadratic weighted kappa (T. Shi et al., 2023); the abstracts of Haque et al. (2022) and Astani and Hasheminejad (2026) likewise report accuracy-based measures only.

Furthermore, a four-bin severity task derived from lesion area has fewer decision

boundaries than a nine-point expert-rated task, those boundaries are defined by a continuous measurable quantity rather than by a rubric, and its ground truth is not subject to inter-rater disagreement in the same way. Results from that literature do not transfer to the present task, and the present task's difficulty should not be judged against its accuracy figures.

Deep Learning for Fall Armyworm

Work specific to FAW is, with one exception noted below, binary or detection-oriented rather than graded. Sena, Pinto, Queiroz, and Viana (2003) established an early image-based approach, segmenting maize plants and classifying image blocks as damaged or non-damaged, with 94.72% of 720 images correctly classified. Ishengoma, Rai, and Said (2021) applied VGG16, VGG19, InceptionV3, and MobileNetV2 to UAV imagery for infected-versus-uninfected classification, with a follow-up hybrid CNN for faster detection (Ishengoma, Rai, & Ngoga, 2022). Zhang et al. (2022) address pest identification rather than damage, distinguishing *S. frugiperda* adults and larvae among 36 maize field pests. Sandhya and Venkataramana (2025) fuse RGB and thermal imagery for infested-versus-healthy classification, and Prabha, Kennedy, Vanitha, Sathiah, and Banu Priya (2021) deploy a MobileNetV2-based CNN in a mobile application that distinguishes healthy from infested plants, reporting 87% validation accuracy. Reported accuracies across these studies range from 87% (Prabha et al., 2021) to 100% (Ishengoma et al., 2021)—figures that reflect the comparative ease of a two-class or identification problem rather than transferable performance on severity grading.

The nearest approach to severity is that of Dzurume et al. (2025), who classify FAW infestation severity in Bangladeshi maize fields into three levels (none, moderate, severe) using temporal Sentinel-2 imagery and a random forest classifier, reporting overall accuracy between 74% and 84%. This is severity work, but at field scale, from satellite multispectral data, with three bins and a non-neural classifier; it addresses neither plant-level assessment, the nine-point scale, nor ordinal structure.

No published work was located that applies CNNs to ordinal grading of FAW foliar damage on the Davis/Prasanna 1–9 scale. FAWCheck v1 applies a CNN to the scale itself, but as nine-class classification (Casasola et al., 2026).

Ordinal Regression for Deep Networks

Niu, Zhou, Wang, Gao, and Hua (2016) introduced the binary-decomposition approach for CNNs—converting a K-class ordinal target into $K-1$ binary "is the rank greater than k?" subtasks—and demonstrated it on age estimation. The approach suffers a structural defect: the independently trained binary classifiers can produce inconsistent predictions, estimating a higher probability that the rank exceeds 2 than that it exceeds 1.

Cao et al. (2020) resolve this with the Consistent Rank Logits (CORAL) framework, in which the $K-1$ binary tasks share a single weight vector and differ only in their bias terms. They prove that this weight sharing forces the predicted cumulative probabilities into non-increasing order, guaranteeing rank consistency by construction rather than by post-hoc correction, and confirm it empirically: zero rank inconsistencies on every dataset and seed, against consistently nonzero counts for the unconstrained predecessor (Cao et al., 2020).

CORAL's structure, a single weight vector shared by all thresholds with a separate bias for each, is the neural counterpart of the proportional-odds model of McCullagh (1980), in which one set of coefficients is shared across all cumulative splits of the scale and only the intercepts differ.

X. Shi et al. (2023) argue that CORAL's guarantee is bought at a cost—the shared weight vector restricts the expressiveness of the output layer—and propose CORN, which achieves rank consistency through conditional training subsets and the chain rule for conditional probabilities, removing the weight-sharing constraint. They report a substantial gain over CORAL. Only in supplementary experiments on two text-review datasets did CORAL attain the lower root mean squared error while CORN attained the

lower MAE, which the authors attribute to RMSE penalizing large errors more heavily; on their image datasets, CORN is lower on both (X. Shi et al., 2023).

A third family abandons binary decomposition and regresses a single scalar score, assigning a class according to where that score falls among class boundaries. The threshold-based ranking loss (THOR) of Fuchs and Keshet (2022) fixes those boundaries in advance at integer values and trains the regression with a pairwise ranking loss, so that inputs are projected onto the segment of their class. Its authors report the best MAE among the ordinal methods they compared, which included CORAL (Fuchs & Keshet, 2022). No official implementation of THOR is published. On five-grade diabetic retinopathy with a ResNet18 backbone, THOR reached MAE 0.709 against CORAL's 0.965 (Fuchs & Keshet, 2022), the most direct published evidence against a rank-consistent head. Plain regression with cut-points tuned after training is the simpler member of the family, but Fuchs and Keshet (2022) include no plain-regression baseline, so their evidence does not rank THOR against it; THOR's fixed boundaries do remove the post-training cut-point step, which would otherwise have to be kept free of evaluation data. Soft ordinal labels (Díaz & Marathe, 2019) encode the ordinal prior in the label distribution rather than the output layer, a separate line of work.

Gutiérrez, Pérez-Ortiz, Sánchez-Monedero, Fernández-Navarro, and Hervás-Martínez (2016) provide the broader taxonomy—naïve approaches, ordinal binary decompositions, and threshold models—across sixteen algorithms and forty-one datasets. The survey predates the CORAL/CORN line and contains no deep learning methods, so it is cited here for its taxonomy and methodological standard rather than for algorithmic guidance.

An important qualification applies to both CORAL and CORN as evidence for the present study. Both were validated on face-age estimation at 33 to 55 ranks. CORN's primary datasets were rebalanced to near-equal class counts before training, and neither paper reports a macro-averaged or otherwise imbalance-aware metric (Cao et al., 2020;

X. Shi et al., 2023). Neither tests small K, and neither tests a distribution with the tail mass of this study's data. THOR's evidence shares the limitation: its five-class experiments did test small K, but the diabetic retinopathy and aesthetics datasets were subsampled to equal class sizes before training (Fuchs & Keshet, 2022). The rank-consistency guarantee is a proof and transfers regardless; the reported magnitude of predictive improvement does not. No method is therefore an expected winner on a nine-level, heavily skewed scale. The one *a priori* consideration is that CORAL's weight-sharing restriction, the expressiveness cost identified by X. Shi et al. (2023), applies to a backbone fine-tuned with each method's own loss, an untested configuration in which that trade-off is most likely to favor CORN.

None of these methods addresses class imbalance in training, and in this study imbalance is handled separately from ordinal structure. Kang et al. (2020) show that a network's learning can be divided into two parts, representation and classifier. They find that imbalance need not harm representation learning, that representations learned on the natural, unrebalanced distribution generalize best, and that strong performance on rare classes can be recovered by adjusting only the classifier. Their classifier retraining (cRT) method keeps the trained representation fixed, re-initializes the classifier, and retrains only the classifier on class-balanced samples (Kang et al., 2020). Because it acts only on the final layer, it applies in the same form to any output layer, which is what this study requires.

Ordinal Methods in Medical Grading

Where agricultural severity work has largely not adopted ordinal methods, medical image grading has. Grading against clinical severity scales is common in that field, and quadratic weighted kappa is its conventional reporting metric even for models trained as ordinary classifiers: a cross-entropy-trained network for diabetic retinopathy reports quadratic weighted kappa of 0.888 (Farag, Fouad, & Abdel-Hamid, 2022), and a cascaded CNN ensemble for five-step AREDS macular degeneration staging reports quadratic kappa

of 0.870 (Xie et al., 2023). Ordinal formulations proper are also established: ordinal regularization for joint diabetic retinopathy and lung nodule grading (Tang, Yang, & Song, 2023), rank-consistent CORAL heads in current retinopathy systems (Bragança et al., 2026), and deep probabilistic ordinal regression for diabetic retinopathy and prostate cancer grading (Toledo-Cortés, Useche, Müller, & González, 2022).

The one located instance of an ordinal severity head in plant imaging is instructive as a contrast case rather than a precedent. Bohra M et al. (2026) propose a multi-task vision transformer for rose leaf disease that includes an ordinal severity head alongside classification and uncertainty estimation. In their limitations, the authors state that the severity labels "originate from disease categories and not from expert-approved measurement systems" (Bohra M et al., 2026), although a later passage describes the annotations as expert-provided; the paper is not internally consistent on this point. Its ablation results, however, settle the practical question: the full model reports severity MAE of 0.000 ± 0.000 with Spearman ρ of 1.000 (Bohra M et al., 2026), indicating that severity is a deterministic function of the class label. The ordinal task is therefore solved trivially and carries no information about ordinal modeling under genuine label uncertainty.

Dataset Bias and Leakage in Plant Imaging

Reported accuracy in plant disease imaging is unusually vulnerable to dataset artifacts. Noyan (2022) demonstrates the mechanism on PlantVillage: a random forest trained on only eight background pixels per image reached 49.0% accuracy across 38 classes, against a 2.6% chance baseline. Mohanty et al. (2016) had earlier reported that their PlantVillage-trained network fell from approximately 99% to approximately 31% accuracy when tested on images from other online sources, a result Noyan (2022) also cites. Background and capture-condition artifacts correlate with class, and a model can learn the correlation instead of the pathology.

This has two direct consequences for the present study. First, the v1 dataset mixes

approximately 1,500 field and client-provided photographs with approximately 584 publicly sourced images, and per-image provenance is not currently recorded (Casasola et al., 2026); if source correlates with severity, a model can learn a dataset signature. Second, the v1 dataset retained multiple images of the same plant where available to increase variation (Casasola et al., 2026), which is precisely the pattern that leaks across randomly assigned splits. Both require explicit control in the evaluation protocol. The size of the leakage effect is measurable: in CIFAR, replacing test images that had near-duplicates in the training set lowered accuracy by 9–14% relative (Barz & Denzler, 2020).

Shift between training and deployment imagery can also be anticipated in training. Training a new output layer on frozen features before fine-tuning keeps a randomly initialized layer from distorting pretrained features, with reported gains of about 1% in distribution and 10% out of distribution (Kumar, Raghunathan, Jones, Ma, & Liang, 2022). Augmenting with the corruptions of the deployment device helps only for those corruptions: networks trained on one distortion become robust to it but generalize poorly to others (Geirhos et al., 2018). JPEG compression and motion blur are among the common corruptions cataloged by Hendrycks and Dietterich (2019).

Evaluation and Comparison Methods for Imbalanced Ordinal Data

Baccianella, Esuli, and Sebastiani (2009) propose macro-averaged variants of MAE, mean squared error, and zero-one error—computing the error separately per true class and then averaging across classes—and prove these coincide with the conventional micro-averaged forms when class cardinalities are equal, while resisting trivial classifiers on imbalanced data. Their diagnostic argument is the substantive one: under micro-averaged MAE on skewed data, the error-minimizing constant prediction need not even be the majority class, so a system that never predicts rare classes can outscore one that attempts them.

The same argument applies to quadratic weighted kappa, which is sensitive to

marginal distributions in the same way. With scores 6 through 9, the susceptible and highly susceptible categories, comprising 14.0% of this dataset, kappa could reward a model that declines to predict severe damage—the specific failure this application cannot tolerate.

Macro-averaging has a countervailing cost the source paper does not address: giving every class equal weight regardless of support means a class with few test items can move the average substantially. Baccianella et al. (2009) validate on one dataset family, one learner, and one right-skewed distribution whose rarest class, though 3.9% of the set, still numbers in the hundreds. Its behavior at the per-class test counts arising in this study is undocumented, which is a reason to measure its stability empirically rather than assume it.

For comparing models, Dietterich (1998) established that paired statistical tests applied across cross-validation folds are anti-conservative, because folds sharing training data are not independent. Nadeau and Bengio (2003) provide the corrected resampled *t*-test, which inflates the variance estimate to account for training-set overlap, and show that a variance estimate that ignores this overlap is too small, making spurious significant differences between algorithms likely. Bouckaert and Frank (2004) found this test replicable, and most replicable when applied to ten-times-repeated ten-fold cross-validation.

Gutiérrez et al. (2016) supply the methodological benchmark this study is held against: nested cross-validation to prevent hyperparameter leakage into the test estimate, twenty to thirty random splits per dataset, and Wilcoxon signed-rank tests with Bonferroni correction. The survey also reports its own external-validity limitation—its discretized-regression datasets are binned into equal-frequency classes and so exhibit no class imbalance, and performance patterns diverge between the discretized and genuinely ordinal halves of its benchmark (Gutiérrez et al., 2016). That is the disclosure the balanced-evaluation choices elsewhere in the ordinal literature needed and did not make.

Reporting Performance Against a Human Ceiling

Where model predictions are compared against expert labels that themselves carry disagreement, agreement among experts bounds what the model can achieve, and reporting it alongside model performance is established practice in clinical imaging. Schaap et al. (2022) provide a clear template: in automating Psoriasis Area and Severity Index scoring, they report the intraclass correlation between CNN and real-life physician scores, report the agreement among five PASI-trained physicians scoring the same images (ICC 0.706–0.793), and compare the CNN's agreement with the real-life scores against the physicians' image-based agreement with them.

None of the ordinal methodology papers reviewed above provides an analog; all treat their ground truth as uncontested. The present study cannot measure an expert ceiling, because it does not have two experts score the same images independently. It adopts a weaker form of the pattern of Schaap et al. (2022): model agreement with an expert is set beside the agreement between the same expert and the author, a rater trained on the rubric, on the same images.

Gaps Addressed by the Present Study

The literature supports four observations that jointly motivate this study.

First, ordinal structure in severity scales is real and consequential, and rank-consistent formulations to exploit it exist and are proven (Cao et al., 2020; X. Shi et al., 2023). Those formulations, however, have been validated at large K on balanced or micro-averaged data, never at nine levels under severe skew.

Second, agricultural severity work has not adopted them. It uses coarse bins derived from measurable area or from class identity, evaluated with plain accuracy (Astani & Hasheminejad, 2026; Bohra M et al., 2026; Haque et al., 2022; T. Shi et al., 2023). FAW-specific work is binary or detection-oriented (Dzurume et al., 2025; Ishengoma et

al., 2022, 2021; Prabha et al., 2021; Sandhya & Venkataramana, 2025; Sena et al., 2003; Zhang et al., 2022), and the one CNN applied to the Prasanna scale treats it as nominal (Casasola et al., 2026).

Third, ordinal regression is used in medical grading against expert-rated scales with measurable rater disagreement (Bragança et al., 2026; Tang et al., 2023; Toledo-Cortés et al., 2022), a field where agreement-based reporting is conventional (Farag et al., 2022; Schaap et al., 2022; Xie et al., 2023). No located work applies rank-consistent ordinal regression to an expert-rated, agronomically standardized damage scale in crop imaging. The Prasanna 1–9 scale sits in that gap.

Fourth, the evaluation standards under which an improvement claim can be defended—stratified leakage-controlled splits, macro-averaged metrics, corrected significance testing, and a measured human ceiling—are established in the methodological and clinical literature (Baccianella et al., 2009; Bouckaert & Frank, 2004; Dietterich, 1998; Gutiérrez et al., 2016; Nadeau & Bengio, 2003; Noyan, 2022; Schaap et al., 2022) but are not yet standard in agricultural computer vision.

This study applies the fourth set of standards to the first set of methods, on a task where neither has previously been brought to bear, with one substitution: an author–expert agreement reference in place of a measured human ceiling.

MATERIALS AND METHODS

This chapter describes the research design, data sources, annotation procedure, evaluation protocol, FAWCheck v1 baseline, models, training procedure, evaluation methods, and statistical analysis used to compare ordinal and multi-class severity models for FAWCheck. Every rule in this chapter, including readouts, the merge rule, the tuning budget, decision rules, and the responses to a failed sensitivity check, is fixed before any ordinal model is trained, and any deviation is reported. The four models compared receive identical folds, backbone, imbalance handling, augmentation, tuning budget, and computing environment.

Research Design

The study follows a comparative experimental design consisting of:

1. **Audit and rebuilding of the evaluation protocol** (see *Evaluation Protocol*).
2. **Expansion and labeling of the dataset**, prioritizing the upper severity range, and **collection of a Philippine external test set** (see *Data Sources*).
3. **Measurement of author–expert agreement** and a **label-consistency check** (see *Expert Annotation, Agreement, and Label Consistency*).
4. **Re-measurement of the current FAWCheck recipe**, ported to PyTorch, to measure the protocol's effect, establish the baseline, and supply the variance for a sensitivity check (see *FAWCheck v1 Baseline and Protocol Effect* and *Sensitivity Check Before the Model Comparison*).
5. **Comparison of softmax classification, CORAL, CORN, and THOR** on MobileNetV3Small, with corrected significance testing (see *Models Compared* and *Evaluation Methods*).

6. **A single final evaluation on the Philippine external test set** (see *Philippine External Test Set*).

Materials

The hardware and software used are listed in Table 2. All runs that enter a single model comparison are made in the same computing environment, and every run records the environment it ran on.

Table 2. Hardware and software

Item	Specification	Role
Primary workstation	ASUS ROG Strix G614JV: Intel Core i7-13650HX (20 threads), 48 GB RAM, NVIDIA GeForce RTX 4060 Laptop GPU (8 GB GDDR6), 2 TB storage, Ubuntu 26.04.1 LTS	All training and evaluation, by default
Fallback	NVIDIA T4 GPU (16 GB GDDR6), through Google Colab	Only when the workstation is unavailable or a run exceeds its GPU memory
Python, PyTorch	—	Every model, including the re-measured v1 baseline
coral-pytorch	Reference implementation by the methods' authors (Raschka, 2022)	CORAL and CORN layers and losses
THOR	Implemented from its paper; no published code (Fuchs & Keshet, 2022)	THOR output layer and loss
Keras (TensorFlow)	Inference only, on the v1 model file	Arm (a) of the protocol-effect design
Albumentations	Buslaev et al. (2020)	Augmentation
scikit-learn	Stratified group *k*-fold splitter	Fold construction
imagehash	64-bit DCT perceptual hash	Near-duplicate clustering

Data Sources

Inherited Labeled Dataset

The primary dataset is the 2,084-image labeled working set assembled for FAWCheck v1, each image assigned an integer score from 1 to 9 against the Prasanna et al. foliar damage descriptors (Casasola et al., 2026; Prasanna et al., 2018). Its composition by severity is given in Table 3. Scores 1 through 4 account for 75.8% of the set, scores 7 through 9 for 7.4%, and score 9 for four images.

Table 3. Severity distribution of the inherited labeled dataset

Score	Images	% of set	Resistance response*
1	385	18.5%	Highly resistant
2	318	15.3%	Resistant
3	487	23.4%	Resistant
4	389	18.7%	Partially resistant
5	214	10.3%	Partially resistant
6	137	6.6%	Susceptible
7	98	4.7%	Susceptible
8	52	2.5%	Highly susceptible
9	4	0.2%	Highly susceptible
Total	2,084	100%	

*Response categories as listed in Table 1.

Its source composition and missing per-image provenance are described under *Dataset Bias and Leakage in Plant Imaging*; provenance is reconstructed before training (see *Leakage and Provenance Controls*).

Expansion Pool

An unlabeled expansion pool is available, comprising the `maize_faw` subset of the YEESI Lab Dataset (Fue et al., 2022)—collected in Morogoro, Tanzania, in August and September 2022, and released under a CC-BY-4.0 license—together with images collected locally by the project team. Not all of the 726 YEESI images are usable for this study, leaving approximately 579. Expansion labeling prioritizes the score 6–9 range, where the inherited dataset is thinnest.

The YEESI imagery differs from the deployment context in cultivar, soil, background vegetation, and ambient light, and 337 of its approximately 579 usable images were captured on an action camera rather than a phone. It is therefore used as training data, not as a validation or test population.

Expansion target. Labeling continues until every score holds at least 100 images in the cross-validation pool, counting inherited and newly labeled images together and excluding the Philippine external test set. Where a score cannot reach 100 from the available sources, its minimum is 15 images, the threshold of the merge rule. On current counts, scores 1 through 6 already exceed 100 (Table 3). Score 7 needs at least 2 more images, score 8 at least 48, and score 9 at least 96, or at least 11 to reach the minimum.

Philippine External Test Set

An external test set of at least 150 images is drawn exclusively from new photographs collected locally by the project team on phones, the deployment device. None comes from the inherited dataset or the YEESI pool, so the external test set is independent of both v1's labeling and the Tanzanian imagery. External test images are labeled through the same pipeline as the expansion pool (see *Labeling Procedure for New Images*). They are reserved before folds are built, excluded from cross-validation and from every model-selection step, and evaluated once, after the final models are trained. No external test image may share

a near-duplicate cluster with any cross-validation image, and a plant photographed more than once contributes to only one side.

The external test set keeps the severity distribution of the photographs as collected rather than being balanced across scores, since its purpose is to estimate performance on the images FAWCheck will actually receive. It therefore supports overall measures and severe-damage recall, with wide intervals at the rare scores, but not per-score claims at scores 8 and 9.

Expert Annotation, Agreement, and Label Consistency

Labeling Procedure for New Images

Every newly labeled image, in the expansion pool and the Philippine external test set, is scored independently by two people against the Prasanna et al. descriptors (Prasanna et al., 2018): the author, and an expert from the IPB Entomology Laboratory. The expert scores each image before seeing the author's score, so that the two scores are independent and free of anchoring. The expert then sees the author's score and assigns the final label, which is the label used for training and evaluation.

Each image's three scores are recorded in an expert correction log: the author's score, the expert's independent score, and the final score, with the expert's reason wherever the final score differs from either. The log distinguishes two kinds of change. First, a *correction* is a final score that differs from the author's. Second, a *revision* is a final score that differs from the expert's own independent score after seeing the author's. Because revisions move the final label toward the author's judgment, their frequency is reported as a direct measure of how much the author's score influenced the final labels.

Photo adequacy. Before scoring, the expert checks three conditions for each new image: the whorl and furl are visible, the image is in focus, and the plant fills enough of the frame for the descriptors to be applied. An image that fails any of them is recorded as

unscorable and excluded from training and evaluation. Unscorable images are counted, and their rate in the external test set is reported as the share of field photographs FAWCheck could not score.

Author–Expert Agreement

Agreement between the author's and the expert's independent scores is computed on all newly labeled images as quadratic weighted kappa, macro-averaged MAE, exact agreement, and within-one-level agreement. It is reported separately for the expansion images and the external test set, since expansion labeling concentrates on scores 6 through 9 while the external test set follows its as-collected distribution, and by score group within each.

This agreement is a human reference point, not a ceiling (see *Scope and Limitations of the Study* and *Human Reference*). Disagreement is expected to concentrate in the mid-to-upper range, where rater reproducibility on this scale is reported to be worst (Srinivasan et al., 2022).

Label Consistency Across Near-Duplicates

The inherited 2,084 labels are not re-scored. Their consistency is checked using the near-duplicate clusters formed for fold construction, which are formed without reference to labels. Images in one cluster are usually photographs of the same plant and should carry the same score. For every cluster of two or more images, this check reports whether its labels disagree, the share of multi-image clusters that do, and the mean absolute difference between paired labels, overall and by score. The same check is applied to the newly labeled images.

The check can reveal inconsistent labeling but cannot establish that labels are accurate: a cluster may hold different plants photographed alike, or one plant whose

damage changed between photographs, and near-duplicates were probably labeled together by the same person. Discordant clusters are reported, not relabeled.

Evaluation Protocol

Why the Inherited Partition Is Discarded

A file-level audit of the inherited partition, conducted by direct count and hashing of the dataset directories rather than from the v1 study's reported figures, finds it non-representative (see Appendix C). Score 1 is 11.7% of training but 47.0% of validation; score 6 has no test images and score 9 none in validation or test; and because validation results guided model selection in v1 (Casasola et al., 2026), checkpoints were selected against a distribution dominated by score 1. These defects lie in how the data were divided, not in the data or their class imbalance, so any comparison computed on this partition measures the partition, and it is discarded rather than patched.

Repeated Stratified Group Cross-Validation

Evaluation uses five-fold cross-validation repeated over three random seeds, for fifteen train-and-evaluate runs. Folds are constructed with stratification on the merged label (see *Pre-Registered Merge Rule*) and grouping on near-duplicate cluster (see *Leakage and Provenance Controls*). Training uses the full nine-level score; the merged label governs fold assignment only.

At $n = 2{,}084$, each fold holds approximately 417 images, including approximately 20 score-7 and 11 score-8 images. These supports are thin enough to require confidence intervals on per-class metrics and substantial enough for those intervals to be meaningful.

No fold's validation data influence training on that fold, including early stopping, checkpoint selection, and any threshold tuning.

Leakage and Provenance Controls

- *Near-duplicate grouping.* The v1 dataset retained multiple images of the same plant (Casasola et al., 2026), which leak across random splits (see *Dataset Bias and Leakage in Plant Imaging*). Near-duplicate cluster is therefore the primary grouping key. Clusters are connected components of a 64-bit perceptual hash at a label-blind calibrated threshold, checked by an embedding-based review of a random sample (see Appendix B, which illustrates clusters in Figure B.1). Each cluster is indivisible in fold assignment.

- *Provenance tagging.* Because source and capture artifacts can correlate with class (Noyan, 2022), each image is tagged with its source—v1 field, v1 public, YEESI, or local collection—and performance is reported by source. If field-photograph performance is materially worse than public-image performance, that gap is the deployment-relevant estimate.

- *Device tagging.* Capture device is recorded, and performance is reported by device. Device is not a grouping key because, where device correlates with severity, device grouping makes stratification impossible and yields folds that measure the device gap rather than model quality.

FAWCheck v1 Baseline and Protocol Effect

Baseline Re-Measurement

The current model is re-measured by retraining its recipe, ported to PyTorch (see *The v1 Recipe and Its Port*), and that figure—not the v1 published figure—is the baseline against which improvement is claimed. The port is read out by argmax, as v1 was (Casasola et al., 2026), with its median readout also reported. Porting keeps the baseline in the same

framework and computing environment as the four models compared (see *Materials*), so a difference between them cannot reflect the framework. However, the baseline is then a reimplementation. Keras and torchvision distribute separately trained ImageNet weights for MobileNetV3Small, each expecting its own input scaling, so the re-measured figure describes v1's *recipe*, not the v1 model file.

The v1 published results are reported for context only in Table 4, with confidence intervals computed for this study from the published counts, since the v1 study reports none (Casasola et al., 2026). The two sets of figures rest on different evaluation sets, the v1 set over-representing severe damage about twofold (see Appendix C), and are not directly comparable.

Table 4. FAWCheck v1 reported performance with recomputed 95% confidence intervals (Wilson score method)

Reported measure	Point estimate	95% CI	Width
CNN exact accuracy (34/68)	50.00%	[38.4%, 61.6%]	23.1 pp
CNN within ± 1 level (52/68)	76.47%	[65.1%, 85.0%]	19.8 pp
System exact agreement (69/83)	83.13%	[73.7%, 89.7%]	16.0 pp
System within ± 1 level (75/83)	90.36%	[82.1%, 95.0%]	12.9 pp

The v1 CNN macro F1 of 0.4105 is not carried forward as a target: its evaluation set held one image at score 9 and two at score 8, too few for a stable per-class F1.

The v1 Recipe and Its Port

The recipe is read from the trained v1 model file, `fawcheck_finetuned.keras`, by parsing its saved configuration without executing it, and from the v1 manuscript (Casasola et al., 2026); Table E.1 (see Appendix E) lists each setting and its PyTorch counterpart. The file does not record five settings: the number of epochs, the batch size, the stopping and checkpoint criterion, any class weighting, and whether a phase training only the output

layer preceded fine-tuning. These are filled with stated defaults, which are replaced if v1's training code becomes available. Since the purpose is to measure v1's recipe as it was, its quirks are reproduced rather than corrected, including a contrast augmentation that only ever increases contrast. The baseline receives v1's four augmentations only; the phone degradations (see *Augmentation*), linear probing before fine-tuning, and classifier retraining belong to this study's training procedure and are not applied to it.

Separating the Protocol from the Model

Specific Objective 4 asks how much of a change in measured performance is due to the evaluation protocol rather than to the model. The port is therefore trained and evaluated four ways, and only adjacent pairs are compared:

- *(a) The v1 model file*, run for inference only, on the inherited test split (Table C.1).
- *(b) The port under the inherited partition:* trained on the inherited training split, with checkpoints selected on the inherited validation split as in v1 (Casasola et al., 2026), and evaluated on the inherited test split, over three seeds.
- *(c) The port under the rebuilt protocol, on the same images:* the 2,084 inherited images only, under the five-fold, three-seed stratified group cross-validation (see *Repeated Stratified Group Cross-Validation*), evaluated out of fold.
- *(d) The port under the rebuilt protocol, on the full cross-validation pool* after expansion. This is the baseline described under *Baseline Re-Measurement*.

The difference between (a) and (b) is the port's fidelity: framework, pretrained checkpoint, the defaults of Table E.1, and seed. The difference between (b) and (c) is the protocol's effect, the quantity Objective 4 asks for. The recipe, framework, and images are the same in both; what differs is how the images are partitioned, whether near-duplicates may straddle training and evaluation, and which images the result is computed on. The

difference between (c) and (d) reflects the added data. Each figure is reported with its cluster-level bootstrap interval, under the same metrics as the other models. No significance test is applied to the (b)–(c) comparison, since (b) rests on a single partition.

Two limits apply to arms (a) and (b). First, they use v1's partition but not v1's evaluation set: the inherited test split holds 146 images, whereas v1 reported its CNN results on 68 (Casasola et al., 2026). Second, the test split has no images at scores 6 and 9 (Table C.1), so macro-averaged MAE under (a) and (b) averages over the other seven scores only.

Models Compared

Comparison Set

Four models are compared (Table 5), all on the current MobileNetV3Small backbone (Howard et al., 2019), so that differences between output layers are not confounded with architecture and the results remain comparable with v1. Softmax multi-class classification is the control and the formulation of the current v1 model (Casasola et al., 2026). Because CORAL and CORN guarantee that the predicted probability that the score exceeds k does not increase with k, their threshold count equals the median of the predicted distribution over scores. Softmax has no readout fixed by its method, so it is read out in two ways from the same trained model: the median, primary, and the argmax, the rule v1 used (Casasola et al., 2026).

The median readout is primary because the median of a predicted distribution minimizes absolute error, the comparison metric, and it puts softmax, CORAL, and CORN on the same statistic; reading softmax by argmax alone would credit ordinal training with a gain due only to the readout. The argmax result shows, at no extra training cost, how much of any difference from v1 is due to the readout alone.

No ordinal head is treated as the expected winner (see *Ordinal Regression for*

Table 5. Models compared, with output layer, loss, and readout

Model	Output layer	Loss	Readout to a score from 1 to 9
Softmax (control)	9 logits	Cross-entropy	Median of the predicted distribution (primary); argmax, as in v1, also reported
CORAL (Cao et al., 2020)	8 threshold logits; one shared weight vector, a bias per threshold	CORAL loss	One plus the number of thresholds k with predicted $P(\text{score} > k) > 0.5$
CORN (X. Shi et al., 2023)	8 conditional logits; no weight sharing	CORN conditional loss	As CORAL, with $P(\text{score} > k)$ the cumulative products of the conditional outputs
THOR (Fuchs & Keshet, 2022)	1 linear unit	Thresholded pairwise ranking loss	Score whose segment $[s-1, s]$ contains the output; outputs beyond the outer boundaries go to 1 or 9

Deep Networks). A finding that ordinal training does not improve severity scoring by a meaningful margin meets the first specific objective, provided it is established by the equivalence test (see *Hypothesis Tests*) rather than inferred from a non-significant difference.

THOR implementation. THOR has no published code and is implemented from its paper's definitions (Fuchs & Keshet, 2022). The paper's boundaries $\{-1, 0, 1, \ldots, K\}$, for classes indexed from zero, become $\{0, 1, \ldots, 9\}$ here, so that score s occupies $[s-1, s]$. The boundaries and the paper's margin $\gamma = 0.5$ are fixed, not tuned, since the other three models have no output-layer hyperparameters. As in the paper, the loss uses pairs of images with adjacent scores: each image in a batch is paired with a randomly drawn training image one score higher, or one lower for score 9, so the anchors keep the natural distribution in stage 1 and the class-balanced distribution in classifier retraining. The output layer is a single linear unit rather than the paper's two added fully connected layers, to keep its capacity comparable with the other three models, and it is trained by this study's procedure (see *Training Procedure*), not the paper's optimizer settings.

Scope of the Comparison, and Why THOR

THOR represents the regression-based family, for the reasons given in *Ordinal Regression for Deep Networks*. No claim is made about THOR against plain regression with tuned cut-points, which its source does not test (Fuchs & Keshet, 2022). Soft ordinal labels (Díaz & Marathe, 2019) are excluded.

Imbalance Handling

Imbalance is handled by classifier retraining (cRT) (Kang et al., 2020), which separates the ordinal-structure effect from the imbalance-handling effect (see *Ordinal Regression for Deep Networks*). It is applied in two stages:

- *Stage 1, representation.* Each model's full network is trained with that model's own loss on the natural, instance-balanced distribution of the training folds, with no rebalancing.
- *Stage 2, classifier retraining.* The backbone is frozen, the model's output layer is re-initialized, and only that layer is retrained on class-balanced samples, restarting the learning rate, for a short schedule (Kang et al., 2020).

Here the output layer is the final layer mapping the backbone's pooled features to the model's outputs (Table 5).

cRT takes the same form for all four models, whereas a per-threshold loss weighting has a natural form only for CORAL and CORN, and it keeps the stage-1 comparison free of reweighting. Hard oversampling is not used, since duplicating images from a 52-image class with known near-duplicates invites memorization.

Pre-Registered Merge Rule

The reporting rule is fixed before expansion-pool labels are collected:

> Any score with fewer than 15 images in the final cross-validation pool is merged with the adjacent score toward the middle of the scale, repeating until every merged group holds at least 15 images. Merged groups are used for fold stratification, for the selection metric, and for headline metrics; per-level results are reported in an appendix.

On current counts, only score 9 ($n = 4$) triggers the rule, forming an {8, 9} group of 56. If the expansion target (see *Expansion Pool*) is met, the merge does not trigger and score 9 enters the headline groups on its own; otherwise the rule applies as a safeguard. Its effect on the metric is described under *Selection Metric*.

Training Procedure

Linear Probing, then Fine-Tuning

Stage 1 of training follows the linear-probe-then-fine-tune (LP-FT) procedure of Kumar et al. (2022). First the pretrained backbone is frozen and only the model's output layer is trained, until validation macro-averaged MAE stops improving. Then the whole network is unfrozen and fine-tuned at a lower learning rate.

This guards against the shift between Tanzanian action-camera training images (see *Expansion Pool*) and Philippine phone photographs at test (see *Dataset Bias and Leakage in Plant Imaging*).

With stage 2 (see *Imbalance Handling*), each model is trained in three phases: linear probing, fine-tuning, and classifier retraining, for every model. Tuning and phase lengths are described under *Hyperparameter Tuning and Compute Budget*.

Model Selection

Early stopping and checkpoint selection use validation macro-averaged MAE, computed on merged groups, with each model's primary readout (see *Comparison Set*), and drawn only from the inner-validation split of each training fold. The argmax readout

of softmax is computed from the checkpoint selected under its median readout, so that the two softmax results describe one trained model.

Augmentation

The augmentation follows v1 (Casasola et al., 2026), extended for phone capture. v1 applied random horizontal flipping, random rotation, random zoom, and random contrast adjustment, to cover variation in leaf orientation, camera angle, distance, and lighting (Casasola et al., 2026); all four are retained.

Three degradations typical of phone photographs are added: JPEG compression, motion blur, and downscaling. They match the deployment device, for the reason given in *Dataset Bias and Leakage in Plant Imaging*, and the phone-only external test set tests whether the robustness was achieved.

The settings are given in Table 6. The geometric settings are those recorded in v1's model file (see *The v1 Recipe and Its Port*). Contrast is varied symmetrically, up and down by 10%, as v1's "random contrast" was evidently intended (the v1 baseline keeps v1's own; see *The v1 Recipe and Its Port*). The lower JPEG quality bound, $q_{\min}$, is the 5th percentile of the JPEG quality estimated from the quantization tables of the locally collected phone photographs, excluding the external test set, so that the trained range matches the deployment device rather than a guess.

JPEG compression is applied last, because compression is the last step of a phone's own image pipeline.

Two augmentations are excluded as not label-preserving, the property Shorten and Khoshgoftaar (2019) call an augmentation's *safety*: vertical flips, which v1 also did not use (Casasola et al., 2026), since whole-plant photographs have a fixed stem-below, leaves-above orientation; and channel shuffling and other color-space transformations, since leaf color may carry information about damage and plant condition (Shorten & Khoshgoftaar, 2019). Contrast adjustment changes contrast without altering hue and is

Table 6. Augmentation settings, applied in the order listed

Transformation	Setting	Probability
Horizontal flip	—	0.5
Rotation	Uniform within $\pm 36°$, reflect padding	1.0
Zoom	Scale 0.9–1.1, aspect ratio preserved, reflect padding	1.0
Contrast	Factor 0.9–1.1 about each channel's mean	1.0
Motion blur	Kernel 3–7 pixels	0.2
Downscaling	Factor 0.5–0.9, then restored to 224×224	0.3
JPEG compression	Quality $q_{\min}$–95	0.5

retained. The pipeline, including its parameters and probabilities, is not tuned per model.

Hyperparameter Tuning and Compute Budget

Inner split. Within each outer training fold, one grouped inner split divides the data about 80:20 into inner-training and inner-validation sets. It uses the same stratified group splitter as the outer folds, so that no near-duplicate cluster straddles the two. The inner-validation set alone governs every data-dependent choice in that fold:

- when linear probing stops;
- early stopping and checkpoint selection in fine-tuning and classifier retraining;
- the choice of hyperparameters.

The outer test fold is used for none of them, which avoids the bias of tuning and evaluating on the same data (Cawley & Talbot, 2010; Raschka, 2018).

The settings are listed in Table 7. Hyperparameters are tuned by random search (Bergstra & Bengio, 2012), and every model receives the same number of search trials, B, since unequal tuning effort can masquerade as a difference between methods (Melis,

Dyer, & Blunsom, 2018). Phase lengths are not searched. In each fold, the best trial on the inner-validation set is that fold's model and is not retrained.

Table 7. Training and tuning settings, identical for all four models

Setting	Value
Phases	Linear probing, fine-tuning, classifier retraining
Inner split	One grouped, stratified 80:20 split of each outer training fold
Search	Random search, B trials per model and fold
Linear-probe learning rate	10^{-4} to 3×10^{-2}, log-uniform
Fine-tuning learning rate	10^{-6} to 10^{-3}, log-uniform
Classifier-retraining learning rate	10^{-4} to 3×10^{-2}, log-uniform
Weight decay	10^{-6} to 10^{-2}, log-uniform
Optimizer; batch size	AdamW; 32 (fixed)
Early stopping	Inner-validation macro-averaged MAE, patience of 8 epochs
Epoch caps	30 (linear probing), 50 (fine-tuning), 20 (classifier retraining)

Compute budget. B is fixed from a compute budget of 56 GPU-hours on the primary workstation, the lower end of the 56–84 hours available.

- Once step 1 of the sensitivity check (see *Sensitivity Check Before the Model Comparison*) has measured the cost of one training, B is set to the largest value for which the projected total fits within 56 hours with a 10% contingency. The total covers the v1 baseline, the primary comparison, and the final models.
- B is written into the study record before any ordinal model is trained.
- Hours beyond 56 are not used to raise B. They go to reruns.

Final Models

Cross-validation estimates the performance of a training procedure. The final model of each method is therefore obtained by applying that same procedure once to the whole cross-validation pool: a grouped inner split of the pool, the same tuning budget B, and the

three training phases with early stopping. This follows the practice of refitting with the selected settings on all available training data after model selection (Raschka, 2018).

Each method's final model is trained with three seeds. The external test result for a method is the mean over its seeds, with their range, so that training variation is separated from external test sampling variation. All methods, including the v1 port, are scored on the external test set once and descriptively. The external test set is never used to choose among them.

Evaluation Methods

Selection Metric

Models are selected on macro-averaged MAE, computed per true class on merged groups and then averaged across classes (Baccianella et al., 2009):

$$\mathrm{MAE}^{M} = \frac{1}{G}\sum_{g}\frac{1}{|g|}\sum_{i:\,y_i \in g}|\hat{y}_i - y_i|,$$

where G is the number of merged groups (nine if no merge triggers) and y_i and $\hat{y}_i$ are the reference and predicted scores on the original 1–9 scale. The inner mean runs over the images whose reference score falls in group g. Merging therefore changes only how classes are weighted: a reference 9 predicted as 8 still counts as an error of one. Macro-averaged mean squared error, MSE^{M}, is defined in the same way with squared errors (Baccianella et al., 2009). Quadratic weighted kappa is deliberately not the selection metric, for the reasons given under *Evaluation and Comparison Methods for Imbalanced Ordinal Data*. Because a thin class can move macro-averaged MAE, its stability is measured through the repeated cross-validation and bootstrap protocol rather than assumed.

Reported Metrics

Reported alongside the selection metric are the measures in Table 8. Every metric is reported for softmax under both its median and argmax readouts, with the median used in the primary comparisons. Results are also broken down by source and by device, with per-class confidence intervals throughout.

Table 8. Metrics reported alongside the selection metric

Metric	Purpose
Quadratic weighted kappa (Cohen, 1968)	Comparability with the medical grading and agricultural literature
Macro-averaged MSE (Baccianella et al., 2009)	Outlier-sensitive measure for Specific Objective 2
Root mean squared error	Comparability with X. Shi et al. (2023)
Plain MAE; within-one-level accuracy	Unweighted summaries of error
Band-crossing error rate; resistance-response confusion matrix	Share of predictions in a different response category from the reference (see *Background of the Study*), since macro-averaged MAE weights band-crossing and within-band errors equally
Severe-damage recall	Share of images with a reference score of 6 or above, the susceptible and highly susceptible categories of Prasanna et al. (2018), that the model also scores at 6 or above. A severe image scored 5 is a miss, because that step crosses into the partially resistant category
Exact accuracy; macro F1	Continuity with v1, subject to the caveats under *Baseline Re-Measurement*
Full confusion matrices	Per level and per merged group

Human Reference

The author's agreement with the expert is reported beside the headline figures. This follows a weaker form of the pattern of Schaap et al. (2022). Each model and the author are compared under the same conditions. Both are scored on exactly the same newly labeled images. Both are also checked against the same reference, which is the expert's independent score. For expansion images in the cross-validation pool, the model's scores

are its out-of-fold predictions, averaged over the three seeds. For external test images, they are the predictions of the final models.

Two figures are reported side by side for each model. The first is the model's macro-averaged MAE against the expert. The second is the author's macro-averaged MAE against the expert on the same images. Each comes with a 95% cluster bootstrap interval (see *Statistical Comparison*), and quadratic weighted kappa is reported with them. The comparison is descriptive, so no margin or verdict is applied. If a model's agreement approaches the author's, the model rates about as consistently as a person trained on the rubric. This does not show that it rates as an expert would (see *Author–Expert Agreement*). Expansion labeling concentrates on scores 6 through 9, so the expansion-image result describes the severe range only.

Statistical Analysis

Statistical Comparison

Because repeated cross-validation folds share training data, a naive paired *t*-test across the fifteen runs is anti-conservative (Dietterich, 1998). Model comparisons therefore use the corrected resampled *t*-test (Nadeau & Bengio, 2003). Bootstrap confidence intervals resample at the level of the near-duplicate cluster rather than the image, since image-level resampling on data containing near-duplicates yields intervals that are too narrow. Because grouped folds vary slightly in size, the test-to-training size ratio in the correction is taken as its mean across folds. Bootstrap intervals use 2,000 resamples and the percentile method. Clusters are resampled within score strata, so that every score is present in every resample, as macro-averaged metrics require.

Sensitivity Check Before the Model Comparison

Before any ordinal model is trained, a sensitivity power analysis (Lakens, 2022) checks whether the design, at its fixed sample size, power, and α, can detect the smallest effect of interest defined below. Its steps, and the responses if it fails, are fixed here.

Smallest effect of interest. The smallest effect of interest is a reduction of 0.10 in macro-averaged MAE, for an ordinal model relative to the median-readout softmax control. That is one fewer one-step error for every ten images in each class. It is set by what would matter in use, not estimated from preliminary runs, since power analyses built on pilot effect sizes are biased (Albers & Lakens, 2018), and it is not revised after any result is seen. The baseline run in step 1 is used only for its variance.

Primary comparisons. The check is computed for the three primary comparisons: CORAL, CORN, and THOR, each against the median-readout softmax control on macro-averaged MAE. Each is assessed at $\alpha' = 0.05/m$ for m primary comparisons, the strictest step of the Holm procedure that governs the tests. Because the variance bound in step 1 applies to any pair of models, the check also covers the secondary comparisons of Specific Objective 2.

Step 1: measure the variance on the current images. The v1 recipe is run on the current 2,084 images under the rebuilt protocol, which is arm (c) of the protocol-effect design (see *Separating the Protocol from the Model*) and so needs no extra training. Its fifteen per-fold macro-averaged MAE values, under the median readout, give the variance from which the smallest effect detectable at 80% power, δ, is computed with the corrected resampled standard error (Nadeau & Bengio, 2003); the derivation is given in Appendix D.

Step 2: project to the planned 2,230 images. The expansion target adds 146 images, all at scores 7–9 (see *Expansion Pool*), so δ is projected class by class to the planned per-fold test counts (see Appendix D). Once expansion is complete, the detectable effect is recomputed from the actual baseline run (arm (d)) and reported with the results, whatever

it shows.

Step 3: decide. The check passes if the projected δ is no larger than the smallest effect of interest. If it passes, the comparison proceeds as planned. If it fails, the following responses are taken in this order, and the first that passes is adopted:

1. *Expand further.* The projection is repeated at larger planned counts to find the expansion that would pass. This response is taken if those labels can be obtained from the expansion pool and local collection within the study's timeline.

2. *Report the ordinal comparison as exploratory.* All four models are still trained and reported with their confidence intervals, but no significance test is reported as confirmatory, and no conclusion is drawn that ordinal training does or does not improve severity scoring. Pre-specifying which analyses are confirmatory keeps hypothesis testing distinct from hypothesis generation (Nosek, Ebersole, DeHaven, & Mellor, 2018). Under this outcome, the rebuilt protocol, its measured effect (Specific Objective 4), and the label-consistency findings become the study's primary contribution.

The check, its inputs, and the response taken are reported in full.

Hypothesis Tests

Two families of one-sided tests are specified (Table 9). Each has familywise $\alpha = 0.05$, controlled by Holm's procedure (Holm, 1979).

- *Family A* carries Specific Objective 1.

- *Family B* carries the two published expectations of Specific Objective 2, in the directions reported by X. Shi et al. (2023) and Fuchs and Keshet (2022). The direction of the MSE test comes from text-review data only; on image data, X. Shi et al. (2023) report CORN ahead on both MAE and RMSE.

Each test uses the corrected resampled *t*-test (see *Statistical Comparison*), with softmax read out by its median.

Table 9. Pre-registered hypothesis tests

Family	Objective	One-sided comparison	Metric
A (primary)	1	CORAL better than softmax	Macro-averaged MAE
A (primary)	1	CORN better than softmax	Macro-averaged MAE
A (primary)	1	THOR better than softmax	Macro-averaged MAE
B (secondary)	2	CORN better than CORAL	Macro-averaged MAE
B (secondary)	2	CORAL better than CORN	Macro-averaged MSE
B (secondary)	2	THOR better than CORAL	Macro-averaged MAE

Each Family A comparison is also given an equivalence test: two one-sided tests against bounds of ± 0.10 macro-averaged MAE, the smallest effect of interest, using the same corrected standard error. Equivalence is shown when the $(1 - 2\alpha')$ interval of the difference lies entirely within the bounds, with α' Holm-adjusted across the family. An equivalence test asks whether an effect is smaller than the smallest effect of interest, which is what allows a null result to be informative (Lakens, 2017; Lakens, Scheel, & Isager, 2018). Each Family A comparison is reported under one of four outcomes:

- significant and not equivalent: the ordinal model is better by a meaningful margin;
- significant and equivalent: it is better, but by less than the smallest effect of interest;
- not significant and equivalent: a meaningful improvement is ruled out;
- not significant and not equivalent: inconclusive.

Family B carries no equivalence test, since the smallest effect of interest is defined on macro-averaged MAE alone. All other comparisons in the study are descriptive: they are reported with estimates and confidence intervals but no significance test.

RESULTS AND DISCUSSION

To be written after the experiments are complete.

SUMMARY AND CONCLUSION

To be written after the experiments are complete.

LITERATURE CITED

ALBERS, C., & LAKENS, D. (2018). When power analyses based on pilot data are biased: Inaccurate effect size estimators and follow-up bias. *Journal of Experimental Social Psychology*, *74*, 187–195. doi: 10.1016/j.jesp.2017.09.004

ASTANI, M., & HASHEMINEJAD, M. (2026). A parallel convolutional neural network with background removal and lesion segmentation for field plant disease severity classification. *Scientific Reports*, *16*, Article 22396. doi: 10.1038/s41598-026-51539-2

BACCIANELLA, S., ESULI, A., & SEBASTIANI, F. (2009). Evaluation measures for ordinal regression. In *Proceedings of the 9th international conference on intelligent systems design and applications (ISDA)* (pp. 283–287). doi: 10.1109/ISDA.2009.230

BARZ, B., & DENZLER, J. (2020). Do we train on test data? Purging CIFAR of near-duplicates. *Journal of Imaging*, *6*(6), Article 41. doi: 10.3390/jimaging6060041

BERGSTRA, J., & BENGIO, Y. (2012). Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, *13*, 281–305.

BOHRA M, N., PATIL, A. R., THORVE, O. H., SHAHADE, A. K., DESHMUKH, P. V., PATIL, S., & KOTECHA, K. (2026). Rose leaf disease classification and severity estimation using an interpretable vision transformer-based multi-task framework. *BMC Plant Biology*, *26*, Article 1026. doi: 10.1186/s12870-026-08786-2

BOUCKAERT, R. R., & FRANK, E. (2004). Evaluating the replicability of significance tests for comparing learning algorithms. In *Advances in knowledge discovery and data mining (PAKDD)* (Vol. 3056, pp. 3–12). doi: 10.1007/978-3-540-24775-3_3

BOULENT, J., FOUCHER, S., THÉAU, J., & ST-CHARLES, P.-L. (2019). Convolutional neural networks for the automatic identification of plant diseases. *Frontiers in Plant Science*, *10*, Article 941. doi: 10.3389/fpls.2019.00941

BRAGANÇA, H., CALIARI, Í. P., VITAL, W. L., FONTENELE, A., CAVALCANTE, S., & MESSIAS, G. (2026). Toward reliable diabetic retinopathy screening. *Sensors*, *26*(14), Article 4515. doi: 10.3390/s26144515

BUSLAEV, A., IGLOVIKOV, V. I., KHVEDCHENYA, E., PARINOV, A., DRUZHININ, M., & KALININ, A. A. (2020). Albumentations: Fast and flexible image augmentations. *Information*, *11*(2), Article 125. doi: 10.3390/info11020125

CAO, W., MIRJALILI, V., & RASCHKA, S. (2020). Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters*, *140*, 325–331. doi: 10.1016/j.patrec.2020.11.008

CASASOLA, A. C., ANACLETO, M. R. U., & NOCUM, J. D. L. (2026). *FAWCheck: A maize damage rating scale* [CMSC 190 Special Problem]. Laguna, Philippines.

CAWLEY, G. C., & TALBOT, N. L. C. (2010). On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, *11*, 2079–2107.

COHEN, J. (1968). Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit. *Psychological Bulletin*, *70*(4), 213–220. doi: 10.1037/h0026256

DAVIS, F. M., NG, S. S., & WILLIAMS, W. P. (1992). *Visual rating scales for screening whorl-stage corn for resistance to fall armyworm* (Technical Bulletin No. 186). Mississippi State, MS: Mississippi Agricultural and Forestry Experiment Station.

DÍAZ, R., & MARATHE, A. (2019). Soft labels for ordinal regression. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)* (pp. 4733–4742). doi: 10.1109/CVPR.2019.00487

DIETTERICH, T. G. (1998). Approximate statistical tests for comparing supervised classification learning algorithms. *Neural Computation*, *10*(7), 1895–1923. doi: 10.1162/089976698300017197

DZURUME, T., DARVISHZADEH, R., DUBE, T., BABU, T. S. A., BILLAH, M., ALAM, S. N., ... NELSON, A. (2025). Detection of fall armyworm infestation in maize fields during vegetative growth stages using temporal Sentinel-2. *International Journal of Applied Earth Observation and Geoinformation*, *139*, Article 104516. doi: 10.1016/j.jag.2025.104516

FARAG, M. M., FOUAD, M., & ABDEL-HAMID, A. T. (2022). Automatic severity classification of diabetic retinopathy based on DenseNet and convolutional block attention module. *IEEE Access*, *10*, 38299–38308. doi: 10.1109/ACCESS.2022.3165193

FERENTINOS, K. P. (2018). Deep learning models for plant disease detection and diagnosis. *Computers and Electronics in Agriculture*, *145*, 311–318. doi: 10.1016/j.compag.2018.01.009

FUCHS, T. S., & KESHET, J. (2022). *THOR: Threshold-based ranking loss for ordinal*

regression. arXiv preprint arXiv:2205.04864.

FUE, K., MASSAWE, D., BARAKABITZE, A., GEOFREY, A., LEBALWA, B., LYIMO, N., ... RAINS, G. (2022). *The YEESI Lab dataset.* Zenodo. doi: 10.5281/zenodo .7893938

GEIRHOS, R., TEMME, C. R. M., RAUBER, J., SCHÜTT, H. H., BETHGE, M., & WICHMANN, F. A. (2018). Generalisation in humans and deep neural networks. In *Advances in neural information processing systems (NeurIPS)* (Vol. 31, pp. 7549–7561).

GUTIÉRREZ, P. A., PÉREZ-ORTIZ, M., SÁNCHEZ-MONEDERO, J., FERNÁNDEZ-NAVARRO, F., & HERVÁS-MARTÍNEZ, C. (2016). Ordinal regression methods: Survey and experimental study. *IEEE Transactions on Knowledge and Data Engineering*, *28*(1), 127–146. doi: 10.1109/TKDE.2015.2457911

HAQUE, M. A., MARWAHA, S., ARORA, A., DEB, C. K., MISRA, T., NIGAM, S., & HOODA, K. S. (2022). A lightweight convolutional neural network for recognition of severity stages of maydis leaf blight disease of maize. *Frontiers in Plant Science*, *13*, Article 1077568. doi: 10.3389/fpls.2022.1077568

HENDRYCKS, D., & DIETTERICH, T. (2019). Benchmarking neural network robustness to common corruptions and perturbations. In *Proceedings of the international conference on learning representations (ICLR).*

HOLM, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, *6*(2), 65–70.

HOWARD, A., SANDLER, M., CHEN, B., WANG, W., CHEN, L.-C., TAN, M., ... LE, Q. V. (2019). Searching for MobileNetV3. In *Proceedings of the IEEE/CVF international conference on computer vision (ICCV)* (pp. 1314–1324). doi: 10.1109/ICCV.2019.00140

ISHENGOMA, F. S., RAI, I. A., & NGOGA, S. R. (2022). Hybrid convolution neural network model for a quicker detection of infested maize plants with fall armyworms using UAV-based images. *Ecological Informatics*, *67*, Article 101502. doi: 10.1016/j.ecoinf.2021.101502

ISHENGOMA, F. S., RAI, I. A., & SAID, R. N. (2021). Identification of maize leaves infected by fall armyworms using UAV-based imagery and convolutional neural networks. *Computers and Electronics in Agriculture*, *184*, Article 106124. doi: 10.1016/j.compag.2021.106124

KANG, B., XIE, S., ROHRBACH, M., YAN, Z., GORDO, A., FENG, J., & KALANTIDIS, Y. (2020). Decoupling representation and classifier for long-tailed recognition. In *Proceedings of the international conference on learning representations (ICLR).*

KUMAR, A., RAGHUNATHAN, A., JONES, R., MA, T., & LIANG, P. (2022). Fine-tuning can distort pretrained features and underperform out-of-distribution. In *Proceedings of the international conference on learning representations (ICLR).*

LAKENS, D. (2017). Equivalence tests: A practical primer for *t* tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, *8*(4), 355–362. doi: 10.1177/1948550617697177

LAKENS, D. (2022). Sample size justification. *Collabra: Psychology*, *8*(1), Article 33267. doi: 10.1525/collabra.33267

LAKENS, D., SCHEEL, A. M., & ISAGER, P. M. (2018). Equivalence testing for psychological research: A tutorial. *Advances in Methods and Practices in Psychological Science*, *1*(2), 259–269. doi: 10.1177/2515245918770963

LU, J., TAN, L., & JIANG, H. (2021). Review on convolutional neural network (CNN) applied to plant leaf disease classification. *Agriculture*, *11*(8), Article 707. doi: 10.3390/agriculture11080707

MCCULLAGH, P. (1980). Regression models for ordinal data. *Journal of the Royal Statistical Society: Series B (Methodological)*, *42*(2), 109–142. doi: 10.1111/j.2517-6161.1980.tb01109.x

MELIS, G., DYER, C., & BLUNSOM, P. (2018). On the state of the art of evaluation in neural language models. In *Proceedings of the international conference on learning representations (ICLR).*

MOHANTY, S. P., HUGHES, D. P., & SALATHÉ, M. (2016). Using deep learning for image-based plant disease detection. *Frontiers in Plant Science*, *7*, Article 1419. doi: 10.3389/fpls.2016.01419

NADEAU, C., & BENGIO, Y. (2003). Inference for the generalization error. *Machine Learning*, *52*(3), 239–281. doi: 10.1023/A:1024068626366

NIU, Z., ZHOU, M., WANG, L., GAO, X., & HUA, G. (2016). Ordinal regression with multiple output CNN for age estimation. In *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 4920–4928). doi: 10.1109/CVPR.2016.532

NOSEK, B. A., EBERSOLE, C. R., DEHAVEN, A. C., & MELLOR, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, *115*(11), 2600–2606. doi: 10.1073/pnas.1708274114

NOYAN, M. A. (2022). *Uncovering bias in the PlantVillage dataset.* arXiv preprint arXiv:2206.04374.

PRABHA, R., KENNEDY, J. S., VANITHA, G., SATHIAH, N., & BANU PRIYA, M. (2021). Artificial intelligence-powered expert system model for identifying fall armyworm infestation in maize (*Zea mays* L.). *Journal of Applied and Natural Science*, *13*(4), 1339–1349. doi: 10.31018/jans.v13i4.3040

PRASANNA, B. M., HUESING, J. E., EDDY, R., & PESCHKE, V. M. (Eds.). (2018). *Fall armyworm in Africa: A guide for integrated pest management*. Mexico City, Mexico: International Maize and Wheat Improvement Center (CIMMYT).

RASCHKA, S. (2018). *Model evaluation, model selection, and algorithm selection in machine learning.* arXiv preprint arXiv:1811.12808.

RASCHKA, S. (2022, July). *coral-pytorch (version 1.4.0) [python package].* `https://github.com/Raschka-research-group/coral-pytorch`.

SANDHYA, P., & VENKATARAMANA, B. (2025). Robust fall army worm detection in maize using multimodal RGB and thermal image fusion. *Scientific Reports*, *16*, Article 464. doi: 10.1038/s41598-025-29784-8

SCHAAP, M. J., CARDOZO, N. J., PATEL, A., DE JONG, E. M. G. J., VAN GINNEKEN, B., & SEYGER, M. M. B. (2022). Image-based automated psoriasis area severity index scoring by convolutional neural networks. *Journal of the European Academy of Dermatology and Venereology*, *36*(1), 68–75. doi: 10.1111/jdv.17711

SENA, D. G., JR., PINTO, F. A. C., QUEIROZ, D. M., & VIANA, P. A. (2003). Fall armyworm damaged maize plant identification using digital images. *Biosystems Engineering*, *85*(4), 449–454. doi: 10.1016/S1537-5110(03)00098-9

SHI, T., LIU, Y., ZHENG, X., HU, K., HUANG, H., LIU, H., & HUANG, H. (2023). Recent advances in plant disease severity assessment using convolutional neural networks. *Scientific Reports*, *13*, Article 2336. doi: 10.1038/s41598-023-29230-7

SHI, X., CAO, W., & RASCHKA, S. (2023). Deep neural networks for rank-consistent ordinal regression based on conditional probabilities. *Pattern Analysis and Applications*, *26*(3), 941–955. doi: 10.1007/s10044-023-01181-9

SHORTEN, C., & KHOSHGOFTAAR, T. M. (2019). A survey on image data augmentation for deep learning. *Journal of Big Data*, *6*, Article 60. doi: 10.1186/s40537-019-0197-0

SIMÉONI, O., VO, H. V., SEITZER, M., BALDASSARRE, F., OQUAB, M., JOSE, C., ... BOJANOWSKI, P. (2025). *DINOv3.* arXiv preprint arXiv:2508.10104.

SRINIVASAN, T., VINOTHKUMAR, B., SHANMUGAM, P. S., BASKARAN, V., ARULKUMAR, G., PRITHIVA, J. N., ... SATHIAH, N. (2022). A new scoring technique for assessing the infestation of maize fall armyworm, *Spodoptera frugiperda* (J.E. Smith). *Madras Agricultural Journal*, *109*. doi: 10.29321/MAJ.10.000674

TANG, W., YANG, Z., & SONG, Y. (2023). Disease-grading networks with ordinal regularization for medical imaging. *Neurocomputing*, *545*, Article 126245. doi: 10.1016/j.neucom.2023.126245

TOEPFER, S., FALLET, P., KAJUGA, J., BAZAGWIRA, D., MUKUNDWA, I. P., SZALAI, M., & TURLINGS, T. C. J. (2021). Streamlining leaf damage rating scales for the fall armyworm on maize. *Journal of Pest Science*, *94*, 1075–1089. doi: 10.1007/s10340-021-01359-2

TOLEDO-CORTÉS, S., USECHE, D. H., MÜLLER, H., & GONZÁLEZ, F. A. (2022). Grading diabetic retinopathy and prostate cancer diagnostic images with deep quantum ordinal regression. *Computers in Biology and Medicine*, *145*, Article 105472. doi: 10.1016/j.compbiomed.2022.105472

XIE, L., VAGHEFI, E., YANG, S., HAN, D., MARSHALL, J., & SQUIRRELL, D. (2023). Automation of macular degeneration classification in the AREDS dataset, using a novel neural network design. *Clinical Ophthalmology*, *17*, 455–469. doi: 10.2147/OPTH.S396537

ZHANG, H., ZHAO, S., SONG, Y., GE, S., LIU, D., YANG, X., & WU, K. (2022). A deep learning and Grad-CAM-based approach for accurate identification of the fall armyworm (*Spodoptera frugiperda*) in maize fields. *Computers and Electronics in Agriculture*, *202*, Article 107440. doi: 10.1016/j.compag.2022.107440

APPENDICES

Appendix A. Example Images for the Prasanna Scale

Figure A.1 shows one image for each score of Table 1. The images come from the training split of the dataset inherited from FAWCheck v1 (see *Inherited Labeled Dataset*) and carry v1's labels. The figure is for illustration purpose only and it does not represent as a guide or reference for image labeling.

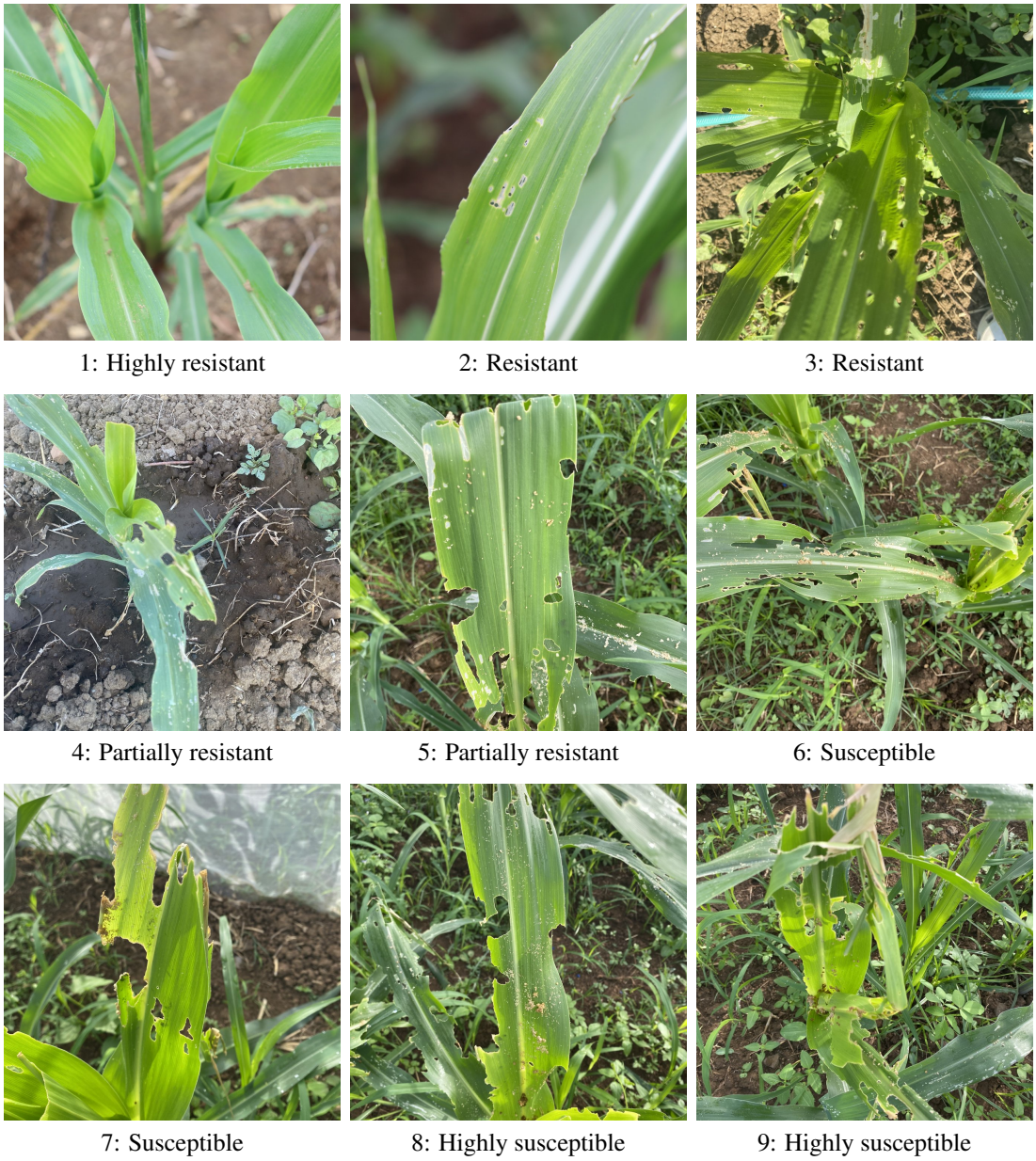


Figure A.1. Example images for scores 1 to 9 of the Prasanna scale

Each panel is a square center crop of one image from the inherited training split, labeled with its score and response (Table 1). Source files, by score: (1) IMG_1442.JPG, (2) IMG_1563.JPG, (3) IMG_5371.jpg, (4) IMG_5091.jpeg, (5) IMG_4770.jpeg, (6) IMG_4766.jpeg, (7) IMG_4528.jpeg, (8) IMG_4518.jpeg, (9) IMG_4486.jpeg.

Appendix B. Near-Duplicate Clustering Procedure

Near-duplicate clusters (see *Leakage and Provenance Controls*) are built in five steps:

- *Hash.* A 64-bit DCT perceptual hash is computed after EXIF orientation, and the smaller Hamming distance of each image and its horizontal mirror is used.
- *Calibrate.* The threshold T is the smallest distance that captures at least 95% of synthetic copies of 200 random images (JPEG quality 70–95, resized to 50–100%, cropped by up to 10%) while admitting under 0.1% of pairs from sources that cannot share a plant, such as v1 field and YEESI images.
- *Inspect.* Pairs within four bits of T are reviewed with labels hidden; T is adjusted once if they show systematic errors, then frozen before folds are built.
- *Cluster.* Clusters are connected components of pairs at distance $\leq T$; any cluster over 20 images is checked for chaining and split where it joins unrelated images.
- *Second pass.* Because hashing misses the same plant photographed from a new angle, each image's five nearest neighbors by cosine similarity of frozen backbone features are reviewed for a random sample of 200 images. A sampled image is a miss if a neighbor is confirmed to be the same plant but lies in a different cluster. If 5 or more of the 200 (2.5%) are misses, the review is extended to every image. Confirmed same-plant pairs are added to the clusters in either case.

The threshold, calibration distributions, inspection counts, cluster sizes, and the second-pass miss count and additions are reported with the dataset.

Figure B.1 shows three clusters from the inherited dataset, with the Hamming distance d of each image's hash from the first image in its row.

(a) The same file saved twice, in two splits under two scores



train, score 7 reference | validation, score 8 $d = 0$

(b) Burst shots of one leaf, split between training and validation



train, score 3 reference | validation, score 5 $d = 4$ | validation, score 5 $d = 10$

(c) A four-shot sequence of one whorl, all in training



train, score 3 reference | train, score 3 $d = 14$ | train, score 5 $d = 8$ | train, score 5 $d = 10$

Figure B.1. Near-duplicate clusters in the inherited dataset

Each panel is a square center crop. Each row holds images of one plant that the inherited partition treats as independent; in (a) and (b) one cluster spans two splits and carries two scores. Distances use the hash of the *Hash* step at an illustrative threshold of 10 bits, not the calibrated T. In (c) the second shot is 14 bits from the first, beyond that threshold, but joins the cluster through the third, which is 8 bits from each: clusters are connected components, not groups of mutually close pairs. Source files: (a) IMG_1689.JPG and its copy; (b) IMG_20190523_135339_5.jpg, _135338_0.jpg, _135338_9.jpg; (c) IMG_1266.JPG to IMG_1269.JPG.

Appendix C. Audit of the Inherited Partition

A file-level audit of the inherited partition, conducted by direct count and hashing of the dataset directories rather than from the v1 study's reported figures, establishes the distribution in Table C.1.

Table C.1. Inherited train/validation/test partition by severity score

Score	Train	Validation	Test	Total
1	186	166	33	385
2	232	45	41	318
3	434	40	13	487
4	324	37	28	389
5	168	25	21	214
6	127	10	0	137
7	78	15	5	98
8	32	15	5	52
9	4	0	0	4
Total	1,585	353	146	2,084

Consequently, the partition has four distinct defects:

- *The validation split is not distributionally representative.* Score 1 is 11.7% of training but 47.0% of validation; score 3 is 27.4% of training but 8.9% of test.
- *Two classes are unevaluable.* Score 6 has 127 training images and none in test, so its test recall is undefined. Score 9 appears only in training and has never been validated or tested.
- *Checkpoint selection was biased.* Because validation results guided model selection in v1 (Casasola et al., 2026), selection was performed against a distribution dominated by score 1—the class with the least operational significance, since it denotes no visible damage.

- *Training and evaluation distributions diverge.* Scores 5–9 constitute 25.8% of the training split but 47.1% of the 68-image set on which the CNN was evaluated (Casasola et al., 2026), over-representing severe damage by approximately twofold.

These defects lie in how the data were divided, not in the data or their class imbalance, so any comparison computed on this partition measures the partition, and it is discarded rather than patched. The v1 study makes no claim that its split was stratified; this is an extension of that work, not a critique.

Appendix D. Pre-Registered Analysis Plan: Derivations

This appendix gives the computations behind steps 1 and 2 of the sensitivity check (see *Sensitivity Check Before the Model Comparison*) and the tuning fallback of the compute budget (see *Hyperparameter Tuning and Compute Budget*).

Step 1: measure the variance on the current images. The v1 recipe is run on the current 2,084 images under the rebuilt protocol, which is arm (c) of the protocol-effect design (see *Separating the Protocol from the Model*) and so needs no extra training. Its fifteen per-fold macro-averaged MAE values, under the median readout, give a variance estimate $\hat{\sigma}^2$. The variance of a per-fold *difference* between two models, unobservable until both are trained, is bounded by $2\hat{\sigma}^2$, its value if their errors were uncorrelated. Models trained on the same folds make positively correlated errors, so the bound is conservative. The corrected standard error of a difference is then

$$\mathrm{SE} = \sqrt{\left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}}\right) \cdot 2\hat{\sigma}^2},$$

with $J = 15$ runs and $n_{\text{test}}/n_{\text{train}}$ the mean ratio across folds (Nadeau & Bengio, 2003), and the smallest detectable effect at 80% power is

$$\delta = \left(t_{1-\alpha'/2,\,J-1} + t_{0.80,\,J-1}\right) \cdot \mathrm{SE}.$$

The baseline's variance, from v1's recipe, stands in for that of the models compared, an approximation reported as such.

Step 2: project to the planned 2,230 images. The expansion target adds 146 images, all at scores 7–9 (see *Expansion Pool*). Macro-averaged MAE weights every class equally, so its variance is dominated by the classes with the fewest test images, which are the classes expansion fills. The projection therefore works class by class. Each class's contribution to $\hat{\sigma}^2$ is scaled by the ratio of its current to its planned per-fold test count, and SE and δ are recomputed. Score 9, now four images and merged with score 8, enters the projection as its own class at its planned count, with the per-image error variance measured for score 8. The scaling treats each class's variance as arising from the size of its test sample. Variation due to training is also reduced by more data, but less predictably. Once expansion is complete, the detectable effect is recomputed from the actual baseline run (arm (d)) and reported with the results, whatever it shows.

Tuning fallback. If the compute budget gives $B < 5$ for the primary comparison, tuning is instead nested in the first seed's five folds only. The configuration chosen most often there is reused for the other two seeds, and the resulting optimistic bias is reported.

Appendix E. FAWCheck v1 Recipe and Its PyTorch Port

Table E.1 lists each setting of the v1 recipe and its PyTorch counterpart (see *The v1 Recipe and Its Port*). Some settings are not recorded in the v1 model file. Their values are stated defaults, and the table marks them as such.

v1's contrast augmentation only ever increases contrast: Keras 3 bounds the lower contrast factor at zero, so a factor of 0.1 yields multipliers between 1.0 and 1.1. The port reproduces this rather than correcting it.

Table E.1. FAWCheck v1 training recipe, as recorded in the v1 model file, and its PyTorch port

Setting	v1 (Keras 3.13.2)	PyTorch port
Backbone	MobileNetV3Small, ImageNet weights, without its classification top	torchvision MobileNetV3-Small, ImageNet weights, classifier removed
Input	224×224 RGB, 0–255, rescaled to $[-1, 1]$ inside the network	224×224 RGB, scaled to $[0, 1]$ and normalized with ImageNet mean and standard deviation
Trainable layers	Last three inverted-residual blocks and final 1×1 convolution; all earlier layers frozen	The same blocks (torchvision `features[9]`–`features[12]`); `features[0]`–`features[8]` frozen
Batch normalization	Frozen layers keep pretrained statistics; momentum 0.999, $\varepsilon = 0.001$	Frozen blocks in evaluation mode; trainable blocks update statistics, momentum 0.001 (PyTorch convention), $\varepsilon = 0.001$
Output layer	Global average pooling $\rightarrow$ dropout 0.3 $\rightarrow$ dense, 9 units, softmax; Glorot-uniform initialization	Global average pooling $\rightarrow$ dropout 0.3 $\rightarrow$ linear, 9 units; Xavier-uniform initialization; softmax at readout
Loss	Sparse categorical cross-entropy	Cross-entropy
Optimizer	Adam, learning rate 1×10^{-5}, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 1 \times 10^{-7}$	Same
Augmentation	Horizontal flip; rotation up to $\pm 36^{\circ}$; zoom up to $\pm 10\%$, aspect ratio preserved; contrast factor between 1.0 and 1.1 per image, about each channel's mean; reflect fill, bilinear interpolation	Same transformations and parameters, applied in the same order after resizing to 224×224
Readout	Argmax	Argmax; median also reported
Class weighting	Not recorded	None (default)
Batch size	Not recorded	32 (default)
Epochs and stopping	Not recorded; checkpoint selection guided by validation results (Casasola et al., 2026)	Up to 50 epochs; early stopping on validation loss with patience 5, best weights restored (default)
Output-layer-only phase before fine-tuning	Not recorded	None (default)